

FUTURE TECH & INFRASTRUCTURE

My Files in the Sky: An Introduction to Cloud Computing

Grade 6 Session: Discover the digital superpower that keeps
your memories safe and accessible everywhere.

1. The Broken Phone Nightmare

Imagine this: You have hundreds of precious photos on your phone—birthday parties, family trips, and funny moments with friends. One day, **splash!** Your phone falls into a puddle. Your heart sinks. Are your memories gone forever?

This is where our digital superpower comes in. What if there was a "magical" place that kept copies of all your memories safe, even when accidents happen? That is exactly what the **Cloud** does.

The Safety Net

The Cloud ensures that your files aren't just stuck on one physical piece of plastic and glass. It saves your precious files safely in the "sky" so you can get them back anytime.

2. It's Not a Weather Cloud!

The first thing to know is that the Cloud is **NOT** a fluffy white object in the sky that brings rain or snow. It is a technological term for something very physical.

The Reality of the Cloud

The Cloud is actually a massive network of **powerful computers** stored in special buildings called **Data Centers**. These computers work together 24/7 to store your photos, videos, and apps.

What it isn't:

Weather, rain, or a literal place in the atmosphere.

What it is:

Buildings full of high-speed computers connected to the internet.

3. Where Does the Cloud Live?

Since the Cloud is made of real computers, it has to live somewhere on Earth. These "digital apartments" are called Data Centers.

Cloud Hubs in India

India is home to many giant data centers. Some of the biggest ones that power your favorite apps are located in:

- **Mumbai:** A major hub for international data.
- **Hyderabad:** Known as a tech capital with massive centers.
- **Chennai:** A key location for undersea cable connections.

These buildings are highly secure, temperature-controlled, and filled with rows upon rows of server racks.

4. Our Digital Bank Locker

To understand why we use the Cloud, think about how your family keeps valuable things like gold or important documents safe.

The "Locker" Comparison

Instead of keeping all your valuables at home where they could be lost in a fire or accident, people put them in a **Bank Locker**. A bank is more secure and has special protection.

The Cloud is your **Digital Bank Locker**. Instead of keeping your data (gold) only on your phone (home), you keep it in a high-security "Digital Vault" that you can access from any "branch" (device) in the world!

Key Benefit: Even if your "home" (phone) is damaged, your "gold" (data) stays safe in the "vault" (Cloud).

5. AWS & Azure: The Landlords

Just like real-world buildings have owners, the digital buildings of the Cloud have owners too. We call them "Digital Landlords."

AWS

Amazon Web Services. They were the pioneers. They own massive complexes that power apps like Netflix and many Indian startups.

Azure

Microsoft Azure. A giant landlord used by many big offices, governments, and schools across India.

These companies build the "apartment complexes" (Infrastructure) and then rent out "rooms" (Storage and Power) to apps you use every day.

6. The Wedding Caterer Secret

One of the most amazing things about the Cloud is something called **Scalability**. Let's use a very Indian example to explain it.

The Wedding Feast

Imagine a wedding caterer. If 50 guests show up, they have enough food. But what if suddenly 500 guests arrive? A great caterer can quickly bring in more stoves, more chefs, and more ingredients to feed everyone.

The Cloud works the same way. If an app like **Zomato** suddenly gets millions of orders during an IPL match, the Cloud "scales up" by instantly providing more computer power to handle the rush. When the match ends, it "scales down" to save energy!

7. Activity: Pack Your Suitcase

Not everything belongs in the Cloud. Some things must stay on your device to work. Can you sort these items correctly?

The Cloud Suitcase (Travels with you)

- Your Emails and Chats
- Photos and Videos
- Saved Game Progress
- Your School Projects (Google Drive/OneDrive)

The Local Device (Stays at home)

- The physical Battery
- The Camera lens
- The Operating System (Windows/Android/iOS)
- The actual Screen

8. The Cloud in Action

You are already using the Cloud every single day! Here are some Indian apps that rely on it:

Hotstar

Uses the Cloud to stream your favorite cricket matches and movies to millions of screens at once.

Paytm

Keeps your digital wallet balance safe in a Cloud "vault" so you can pay from any shop.

Anywhere, Anytime: Because of the Cloud, you can pause a movie on your TV and finish it on your phone while traveling in the car. The Cloud remembers exactly where you stopped!

9. Your Digital Superpower

Now you know that the Cloud is a network of powerful remote computers that keep your life connected.

Session Recap

1. The Cloud is made of real computers in **Data Centers**.
2. **AWS** and **Azure** are the big digital landlords.
3. It keeps your files **safe** and **accessible** from anywhere.
4. It **scales** like a wedding caterer to handle big crowds.

Ready to explore the Cloud further?

Your journey into future technology has just begun! Stay curious.